

Jingbo Wan

Academic Curriculum Vitae

CNRS Postdoctoral Researcher
Sorbonne Université (LJLL)

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Employment

- **CNRS Postdoctoral Researcher**, Sorbonne Université (LJLL), Paris 2025–present
Supervisor: J       Szeftel

Education

- **Ph.D. in Mathematics** Aug 2020 – May 2025
Advisors: Elena Giorgi and Mu-Tao Wang Columbia University
- **B.Sc. in Mathematics (International Research Enrichment Track)** Aug 2016 – May 2020
Indep. studies with: Beifang Chen, Frederick Fong and Hai Zhang HKUST
First-Class Honors
Academic Achievement Medal (CGA: 4.069/4.3)

Research Interests

General Relativity, Partial Differential Equations, Differential Geometry, Geometric Flows

Publications and Preprints

1. (Joint with Jianchun Chu and Man-Chun Lee) *Rigidity of positive mass theorem with fast metric decay* arXiv preprint arXiv:2607.17236 (2026).
2. (Joint with Tsz-Kiu Aaron Chow) *Maximal Normal Curvature and Veronese Rigidity*, arXiv preprint arXiv:2607.00949 (2026).
3. (Joint with Allen Juntao Fang and Elena Giorgi) *Wave Decay and Horizon Instability on Strongly Charged Extremal Kerr–Newman Black Holes*, arXiv preprint arXiv:2606.29956 (2026).
4. (Joint with Tsz-Kiu Aaron Chow) *Sharp Focal Radius Estimate and Rigidity of Hypersurfaces in Manifolds with Positive Curvature*, arXiv preprint arXiv:2606.02829 (2026).
5. (Joint with Dawei Shen) *Vacuum Initial Data with Minimal Decay and Borderline Decay*, arXiv preprint arXiv:2602.01557 (2026).
6. (Joint with Dawei Shen) *Cauchy Data for Formation of Multiple Black Holes with Prescribed ADM Parameters*, arXiv preprint arXiv:2601.01517 (2026).

7. (Joint with Elena Giorgi and Dawei Shen) *Cauchy Data for Multiple Collapsing Boson Stars*, arXiv preprint arXiv:2512.01844 (2025).
8. (Joint with Chung-Jun Tsai, Mao-Pei Tsui, and Mu-Tao Wang) *Constructing Entire Minimal Graphs by Evolving Planes*, *Calc. Var. Partial Differ. Equ.* **65** (2026), article number 196.
9. (Joint with Allen Juntao Fang and Elena Giorgi) *Einstein–Maxwell Equations on Mass-Centered GCM Hypersurfaces*, arXiv preprint arXiv:2510.10814 (2025).
10. (Joint with Allen Juntao Fang and Elena Giorgi) *Mass-Centered GCM Framework in Perturbations of Kerr–Newman*, arXiv preprint arXiv:2510.10811 (2025).
11. (Joint with Dawei Shen) *Formation of Multiple Black Holes from Cauchy Data*, arXiv preprint arXiv:2506.16117 (2025).
12. (Joint with Dawei Shen) *Formation of Trapped Surfaces for the Einstein–Maxwell–Charged Scalar Field System*, arXiv preprint arXiv:2504.19976 (2025).
13. (Joint with Sergiu Klainerman and Dawei Shen) *A Canonical Foliation on Null Infinity in Perturbations of Kerr*, arXiv preprint arXiv:2412.20119 (2024), submitted.
14. (Joint with Elena Giorgi) *Boundedness and Decay for the Teukolsky System in Kerr–Newman Spacetime II*, *Adv. Theor. Math. Phys.* **30** (2026), no. 2, 295–333.
15. *Sharp Interior Gradient Estimate for Area-Decreasing Graphical Mean Curvature Flow in Arbitrary Codimension*, arXiv preprint arXiv:2407.05602 (2024), submitted.
16. (Joint with Man-Chun Lee and Luen-Fai Tam) *Rigidity of Area Non-Increasing Maps*, *Trans. Amer. Math. Soc.* **378** (2025).
17. (Joint with Man-Chun Lee) *Rigidity of Contracting Maps via Harmonic Map Heat Flow*, *Proc. Amer. Math. Soc.* **154** (2026), 3107–3118.
18. (Joint with Elena Giorgi) *Physical-Space Estimates for Axisymmetric Waves on Extremal Kerr*, *J. Funct. Anal.* **287** (2024), no. 12, 110668.
19. (Joint with Florian Johne and Tsz-Kiu Aaron Chow) *Preserving Positive Intermediate Curvature*, *J. Geom. Anal.* **33** (2023), 366.
20. (Joint with Nicholas Cheng-Hoong Chin and Frederick Tsz-Ho Fong) *Uniqueness of Self-Conformal Solutions to Inverse Curvature Flows*, *Proc. Amer. Math. Soc.* **148** (2020), 4967–4982.

Invited Talks

LJLL Seminar on Mathematical General Relativity	Sorbonne
<i>Topic: Wave decay on slowly-rotating extremal Kerr–Newman black holes</i>	04/2026
Mathematical GR and Hyperbolic PDE Seminar	Virtual
<i>Topic: Vacuum asymptotically flat initial data at minimal and borderline decay</i>	04/2026
AMS Congress	Boston College
<i>Topic: Cauchy Data for Formation of Multiple Black Holes with Prescribed ADM Parameters</i>	03/2026
Princeton Analysis Seminar	Princeton
<i>Topic: Constructing Cauchy Data for Multiple Black Hole Formation</i>	03/2026

TSIMF Sanya Waves 2026	Sanya
<i>Topic: Formation of Multiple Black Holes from Cauchy Data</i>	02/2026
Department of Mathematics Seminar	HKUST
<i>Topic: Entire minimal graphs from an evolving-plane ansatz</i>	02/2026
Department of Mathematics Seminar	HKUST
<i>Topic: Vacuum asymptotically flat initial data at minimal and borderline decay via conical solution operators</i>	02/2026
ICCM 2025	Shanghai
<i>Topic: Formation of Multiple Black Holes from Cauchy Data</i>	01/2026
GAPDE Seminar	Cambridge
<i>Topic: Formation of Multiple Black Holes from Cauchy Data</i>	11/2025
Analysis Seminar – AMSS	Virtual
<i>Topic: Hyperbolic Estimates for Nonlinear Stability of Subextremal Reissner–Nordström</i>	04/2025
Hyperbolic & Dispersive Equations on Curved Geometries – SCGP	New York
<i>Topic: Hyperbolic Estimates for Nonlinear Stability of Subextremal Reissner–Nordström</i>	04/2025
Lehigh Geometry Seminar	Virtual
<i>Topic: Rigidity of Contracting Maps Between Positively Curved Manifolds</i>	01/2025
Mathematical GR and Hyperbolic PDE Seminar	Virtual
<i>Topic: Axisymmetric Solution of the Teukolsky System</i>	09/2024
Harmonic Analysis and Differential Equations Seminar	UC Berkeley
<i>Topic: Axisymmetric Solution of the Teukolsky System</i>	09/2024
Mini-Workshop in Geometric Analysis	SUSTech
<i>Topic: Sharp Interior Gradient Estimate for GMCF in Arbitrary Codimension</i>	08/2024
Seminar on Pure Mathematics	HKUST
<i>Topic: Sharp Interior Gradient Estimate for GMCF in Arbitrary Codimension</i>	08/2024
Topics in General Relativity	University of Münster
<i>Topic: Axisymmetric Solution of the Teukolsky System</i>	07/2024
Geometric Analysis Seminar	Westlake University
<i>Topic: Rigidity of Area Non-Increasing Maps</i>	04/2024
Geometric Analysis Seminar	BIMCR
<i>Topic: Rigidity of Area Non-Increasing Maps</i>	03/2024
Geometric Analysis Seminar	CUNY Graduate Center
<i>Topic: Rigidity of Contracting Maps Using Harmonic Map Heat Flow</i>	09/2023
Seminar on Pure Mathematics	HKUST
<i>Topic: Rigidity of Contracting Maps Using Harmonic Map Heat Flow</i>	08/2023
MIST Workshop on Geometry	CUHK
<i>Topic: Rigidity of Contracting Maps Using Harmonic Map Heat Flow</i>	08/2023
Joint Geometric Analysis Seminar	CUHK
<i>Topic: Wave Equations on Black Holes</i>	06/2023
General Relativity & Geometric Analysis Seminar	Columbia University
<i>Topic: The Existence of a Black Hole Due to Condensation of Matter</i>	10/2022

Short Courses

University of Science and Technology of China	USTC
<i>An Introduction to the Geometry of Null Hypersurfaces in General Relativity</i>	06/2026
The Chinese University of Hong Kong	CUHK
<i>On the Formation of Trapped Surfaces</i>	08/2025
The Chinese University of Hong Kong	CUHK
<i>Semi-linear Wave Equation on Black Holes</i>	08/2023

Co-Organized Seminars

Informal Reading Seminar in GR	Columbia University
<i>Topic: various aspects of general relativity</i>	2024 Fall
Student PDE Seminar	Columbia University
<i>Topic: BV Functions and Regularity of the Minimal Surface</i>	2024 Fall
Student Geometric Analysis Seminar	Columbia University
<i>Topic: Regularity Theory of the Monge-Ampère Equation</i>	2024 Spring
Student Geometric Analysis Seminar	Columbia University
<i>Topic: Geometric Flows and Their Applications</i>	2023 Fall
Student Geometric Analysis Seminar	Columbia University
<i>Topic: Scalar Curvature and Dihedral Rigidity</i>	2023 Spring
Student Geometric Analysis Seminar	Columbia University
<i>Topic: Scalar Curvature and Topology</i>	2022 fall
Informal student GR seminar, over zoom	Columbia University
<i>Topic: Semilinear wave equation on blackholes</i>	2021 fall

Summer Schools & Workshops

Introductory Workshop: Special Geometric Structures and Analysis	Berkeley
<i>Program Workshop, SLMath</i>	Sep 3 – Sep 6, 2024
Introductory Workshop: New Frontiers in Curvature	Berkeley
<i>Program Workshop, SLMath</i>	Aug 26 – Aug 30, 2024
Geometric Analysis Workshop	Shenzhen
<i>Mini Workshop, SUSTech</i>	Aug 18 – Aug 19, 2024
Mathematics of General Relativity and Fluids	Crete
<i>Summer Graduate School, SLMath and FORTH</i>	Jul 22 – Aug 2, 2024
Symmetries and Dynamics of Extremal Black Holes	Princeton
<i>Workshop, PCTS, PGI</i>	Mar 6 – Mar 8, 2024
The Fifth Taiwan International Conference on Geometry	Taipei
<i>Mini courses & Conference, NTU</i>	Jan 12 – Jan 19, 2024
Mass, Einstein Constraint Equations, and Penrose Inequality	Stony Brook
<i>Conference, Simon Center</i>	Sep 18 – Sep 22, 2023
Nonlinear Aspects of General Relativity	Princeton
<i>Workshop, PCTS, PGI</i>	Oct 10 – Oct 13, 2023
Conference on Nonlinear Waves and Mathematical Relativity	Beijing
<i>Conference, Tsinghua</i>	Jul 10 – Jul 14, 2023
Rutgers Geometric Analysis Conference	Rutgers
<i>Mini Workshop, Rutgers</i>	May 15 – May 16, 2023
37th Geometry Festival	Princeton
<i>Workshop, Princeton</i>	Apr 28 – Apr 30, 2023
Northeast Workshop in Geometric Analysis (NEWGA)	Connecticut
<i>Mini Workshop, UConn</i>	Nov 5 – Nov 6, 2022

Cauchy Problem in General Relativity
EWM-EMS Summer School, Mittag-Leffler
Geometry of Submanifolds
Workshop, Tianyuan Center of Southeast China

Online
Jun 13 – Jun 17, 2022
 Online
Jul 20 – Jul 24, 2020

Awards & Honors

Dean's Fellow <i>Graduate School of Arts and Sciences, Columbia University</i>	2020-2025
Academic Achievement Medal <i>Department of Mathematics, Hong Kong University of Science and Technology</i>	2020
Chern Class Achievement Scholarship <i>Department of Mathematics, Hong Kong University of Science and Technology</i>	2020
Chern Class Talent Scholarship <i>Department of Mathematics, Hong Kong University of Science and Technology</i>	2017-2020
First Runner-Up, Mr Armin and Mrs Lillian Kitchell Undergraduate Research Award <i>Undergraduate research competition held by Hong Kong University of Science and Technology</i>	2019
Epsilon Fund Award <i>Department of Mathematics, Hong Kong University of Science and Technology</i>	2019
University's Scholarship Scheme for Continuing Undergraduate Students <i>Hong Kong University of Science and Technology</i>	2017-2020
Dean's List <i>Hong Kong University of Science and Technology</i>	2016-2020
32th Chinese Physics Olympiad - First prize <i>within Guangdong Province</i>	2015
Pan Pearl River Delta Physics Olympiad - second prize <i>Hong Kong University of Science and Technology</i>	2014

Teaching

Spring 2025 Calculus II, Instructor
 Fall 2024 Accelerated Multivariable Calculus, Teaching Assistant
 Summer 2024 Analysis and Optimization, Instructor
 Spring 2024 Accelerated Multivariable Calculus, Teaching Assistant
 Fall 2023 Accelerated Multivariable Calculus, Teaching Assistant for second half of semester
 Fall 2023 Number theory and Cryptography, Teaching Assistant for first half of semester
 Spring 2023 Linear Algebra, Teaching Assistant
 Fall 2022 Linear Algebra, Teaching Assistant
 Spring 2022 Number theory and Cryptography, Teaching Assistant
 Fall 2021 Ordinary Differential Equations, Teaching Assistant